

LESSON 3: LOGIC & REASONING

So far, we've seen that AI can learn from data and predict outcomes. But there is more to intelligence — reasoning. We use reasoning all the time. Much of our everyday thinking is reasoning. It helps us understand what we observe, solve problems, and decide what to do.

Reasoning, though we may not always notice it, is structured. There are rules to thinking that guide us toward conclusions. These conclusions, or outcomes of reasoning, may be an understanding, a new insight, a decision, or a plan. The rules that make reasoning work are referred to as logic.

In this lesson, we will explore reasoning, its forms, and the framework in which it operates: logic.

REASONING

Reasoning is a structured way of thinking — or processing — that helps us uncover new information, make decisions, and make sense of our world. In every case, reasoning follows a simple path: input, processing, and output. It begins with what is already known — information or knowledge that serves as the input. This is then processed, and the outcome is produced. In general, three outcomes are possible: an inference, a decision, or an explanation.

REASONING IS THE STRUCTURED WAY WE THINK TO DISCOVER,
DECIDE, OR UNDERSTAND.



INFERENCE

Inference means arriving at a new piece of information. It is the process of linking information we already have in order to reach a conclusion. The information we start with forms the basis or starting point of the reasoning process: these are called premises.

Premises may come from three sources:

1. facts we already know, which form general rules.
2. observations our senses are picking up.
3. predictions based on learned patterns.

A PREMISE IS THE STARTING POINT OF INFERENCE AND CAN BE A FACT, OBSERVATION, OR PREDICTION.

EXAMPLE 3.1 (HI - HUMAN INTELLIGENCE)

Layla is in science class, listening carefully as her teacher explains that all birds have feathers. A moment later, the teacher holds up a picture of a chicken and asks the class, “Is this a bird?” Layla thinks it through. She recalls what she has just learned: all birds have feathers. She observes that chickens have feathers. From this, she reasons that chickens must be birds.

LOGICAL FORM

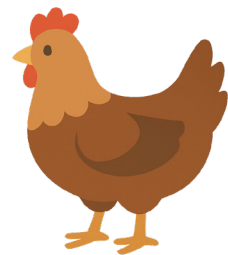
PREMISES

GENERAL RULE: ALL BIRDS HAVE FEATHERS.

OBSERVATION: CHICKENS HAVE FEATHERS.

INFERENCE

THEREFORE : CHICKENS ARE BIRDS.



INFERENCE IS THE PROCESS OF ARRIVING AT NEW INFORMATION.

EXAMPLE 3.1 (HI - ARTIFICIAL INTELLIGENCE)

A video app observes that a student has recently watched several basketball highlights. The AI model behind the app has learned a general rule from past data: *viewers who watch videos about a topic will probably enjoy other videos on that same topic.*

Using this knowledge, the system reasons that the student is likely interested in basketball more broadly. From this, it infers that the student would also enjoy basketball training videos and recommends them.



LOGICAL FORM

PREMISES

GENERAL RULE : VIEWERS WHO WATCH VIDEOS ABOUT A TOPIC WILL PROBABLY ENJOY OTHER VIDEOS ON THAT SAME TOPIC.

OBSERVATION : THE STUDENT WATCHED BASKETBALL HIGHLIGHTS.

INFERENCE

THEREFORE : PROBABLY, THE STUDENT WOULD ENJOY BASKETBALL TRAINING VIDEOS.

Both examples show how inference arrives at a new conclusion by linking what we already know. In the first case, the conclusion “*Chickens are birds*” followed from a general rule (*All birds have feathers*) combined with a specific observation (*Chickens have feathers*). Because this reasoning was grounded in a fact, the inference carried a high degree of certainty.

In the second case, the conclusion “*The student would probably enjoy basketball training videos*” followed from a general rule learned from data (*Viewers who watch videos on a topic will probably enjoy more videos on that topic*) and an observation (*The student watched basketball highlights*). Here the inference relied on a prediction, and since predictions come with uncertainty, the conclusion is expressed only as probable.

THE CERTAINTY OF AN INFERENCE DEPENDS ON THE CERTAINTY OF ITS PREMISES.

The certainty of an inference depends on the certainty of its premises. If the premises are facts, the conclusion is highly certain. If the premises are predictions, the conclusion is less certain and best understood as probably true rather than definitely true.

DECISION-MAKING

Decision-making means selecting the best choice from several possibilities. It is the process of looking at different choices, considering what will probably happen if we take each one, and then selecting the option with the highest utility — the one that is most useful, valuable, or best fits our goal.

DECISION-MAKING IS THE PROCESS OF SELECTING THE CHOICE WITH THE HIGHEST UTILITY.

Decisions are also guided by premises: facts, rules, observations, predictions, and even personal preferences or opinions. Which choice seems best depends on the context — sometimes facts dominate, other times preferences or predictions play the larger role.

EXAMPLE 3.2 (HI - HUMAN INTELLIGENCE)

Lina was getting ready for school and saw that the sky was cloudy. She thought about walking, riding her bike, or asking her father for a ride. She remembered that she didn't mind a little rain and that she really disliked depending on her father's schedule since he sometimes left very early. After weighing her options, she decided to take her bike.



LOGICAL FORM

PREMISES

PREFERENCE: I DON'T MIND A LITTLE RAIN.

PREDICTION: IT IS LIKELY THAT MY FATHER WILL LEAVE VERY EARLY.

CHOICES

CHOICE 1: IF I WALK, THEN PROBABLY I WILL BE SLOW AND MIGHT GET WET.

CHOICE 2: IF I BIKE, THEN PROBABLY I WILL ARRIVE FASTER AND RISK ONLY A LITTLE RAIN.

CHOICE 3: IF I RIDE WITH MY FATHER, THEN PROBABLY I WILL STAY DRY BUT ARRIVE TOO EARLY AT SCHOOL.

DECISION

THEREFORE: I WILL RIDE MY BIKE TO SCHOOL.

In Lina's case, her premises included a personal preference — *she didn't mind a little rain* — and a prediction — *her father will likely leave very early*. These premises shaped how she evaluated her choices.

She compared three choices: *walking*, *biking*, or *riding with her father*. Each choice came with possible outcomes, and she weighed the utility (how useful or valuable each outcome was to her).

1. Walking would take too long and might make her wet (low utility).

2. Riding with her father would keep her dry but could get her to school too early (low utility).
3. Biking was fast, comfortable enough, and matched her preference about rain (highest utility).

Her decision to ride her bike was based on comparing the outcomes against her premises and selecting the choice with the best overall value.

EXAMPLE 3.2 (AI - ARTIFICIAL INTELLIGENCE)

A self-driving car approaches an intersection. Its AI system considers three options: keep driving, slow down, or stop. Its sensors show that a pedestrian is stepping onto the crosswalk and a car is close behind. The car's decision-making is guided by two fundamental rules: never harm humans, and avoid collisions unless it contradicts the first rule. Based on this, the car decides to stop for safety.

LOGICAL FORM

PREMISES

RULE 1: NEVER HARM HUMANS.

RULE 2: AVOID COLLISIONS UNLESS IT CONTRADICTS RULE 1.

CHOICES

CHOICE 1: IF THE CAR KEEPS DRIVING, THEN THE PEDESTRIAN WILL BE HARMED.

CHOICE 2: IF THE CAR SLOWS DOWN, THEN PROBABLY IT WILL AVOID HARM BUT MAY RISK A REAR COLLISION.

CHOICE 3: IF THE CAR STOPS, THEN IT WILL NOT HARM THE PEDESTRIAN, BUT A REAR COLLISION WILL OCCUR.

DECISION

THEREFORE: STOP THE CAR.



For the self-driving car, its premises were programmed as rules: *never harm humans* and *avoid collisions unless that contradicts the first rule*. With these rules in place, the car evaluated three choices: *keep driving*, *slow down*, or *stop*. Each choice had different outcomes.

1. Continuing to drive would harm a pedestrian (very low utility).
2. Slowing down might possibly avoid harm but still risk a probable rear collision (medium utility).
3. Stopping would certainly keep the pedestrian safe even though it might cause a rear collision (highest utility, because protecting humans is the top rule).

The AI system was deciding under a “Definitely Do Not” condition set by its first rule. This condition gave that rule the highest utility or value, so it overruled the other possibilities. This shows how the car compared the possible outcomes against its guiding premises, then selected the option with the greatest utility — the one most consistent with its goals and priorities.

UTILITY IS A MEASURE OF HOW USEFUL, VALUABLE, OR GOAL-FITTING AN OUTCOME IS.

Although Lina and the self-driving car are very different, they both used the same reasoning pattern. Each began with premises — Lina’s were based on her preferences and a prediction, while the car’s were based on safety rules.

They considered several choices, weighed the possible outcomes, and compared the utility of each option. Lina’s decision was shaped by comfort, while the car’s was shaped by safety, but the structure of their reasoning was the same.

THE UTILITY OF A CHOICE DEPENDS ON THE PREMISES USED TO JUDGE IT.

EXPLANATION

Explanation is the process of reasoning backward from a given — something already accepted as true — in order to uncover its causes, contributing conditions, or relationships. In other words, explanation begins with an outcome and asks: “What conditions could have produced this?” Instead of starting with premises and moving forward, as in inference, explanation works in reverse — from an observation to possible reasons. In terms of directionality, explanation is therefore the opposite of inference.

A GIVEN IS THE STARTING POINT OF EXPLANATION, FROM WHICH WE REASON BACKWARD TO UNCOVER ITS CAUSES, CONTRIBUTORS, AND RELATIONSHIPS.

In terms of strategy, however, explanation is closer to decision-making: both involve a search process. Decision-making searches among possible actions to select the one with the highest utility, while explanation searches among possible reasons to identify the one with the greatest plausibility.

THE PLAUSIBILITY OF AN EXPLANATION DEPENDS ON THE PREMISES THAT SUPPORT IT.

Like inference and decision-making, explanation depends on a knowledge base — the rules, facts, or learned patterns that connect conditions to outcomes. Without this background knowledge, reasoning backward would not be possible. But unlike inference, which can sometimes deliver definite conclusions, explanation typically carries varying degrees of uncertainty. The strength of an explanation depends on the plausibility of its premises and how well they fit the observed outcome.

EXAMPLE 3.3 (HI - HUMAN INTELLIGENCE)

Omar steps outside one morning and notices that the front lawn is wet. He wonders what caused it. Two possibilities come to mind: maybe it rained during the night, or maybe the sprinkler was on. Omar thinks about what usually happens: when it rains, everything in the neighborhood gets wet, not just his lawn. But today, he notices that only the front lawn is wet while the rest of the ground is dry. Based on this observation, Omar reasons that the most likely cause is that the sprinkler was on.

LOGICAL FORM

GIVEN

OBSERVATION: THE FRONT LAWN IS WET.

PREMISES

GENERAL RULE 1: IF IT RAINS THEN EVERYTHING IS WET.

GENERAL RULE 2: IF THE SPRINKLER IS ON THEN ONLY THE FRONT LAWN GETS WET.

OPTIONS

OPTION 1: THE CAUSE MIGHT BE RAIN.

OPTION 2: THE CAUSE MIGHT BE THE SPRINKLER.

OBSERVATION: ONLY THE FRONT LAWN IS WET.

EXPLANATION

THEREFORE: SINCE ONLY THE FRONT LAWN IS WET, THE MOST PROBABLE EXPLANATION IS THE SPRINKLER WAS ON.



This example shows how explanation works by reasoning backward from an outcome. Omar begins with what is given — the lawn is wet — and considers different possible reasons. By comparing these options against additional evidence, he chooses the cause that best fits.

EXAMPLE 3.3 (AI - ARTIFICIAL INTELLIGENCE)

A medical AI system is fed patient information: the person has both a fever and a cough. The system compares this data to patterns in its knowledge base. It “knows” that the flu typically produces both fever and cough, while allergies can produce cough but not fever. Since the patient’s data includes fever along with cough, the AI rules out allergies as the cause and identifies the flu as the most probable explanation.



LOGICAL FORM

GIVEN

OBSERVATION: THE PATIENT HAS A FEVER AND COUGH.

PREMISES

GENERAL RULE 1: IF A PATIENT HAS THE FLU, THEN FEVER AND COUGH OCCUR TOGETHER.

GENERAL RULE 2: IF A PATIENT HAS ALLERGIES, THEN COUGH OCCURS BUT FEVER DOES NOT.

OPTIONS

OPTION 1: THE CAUSE MIGHT BE FLU.

OPTION 2: THE CAUSE MIGHT BE ALLERGIES.

OBSERVATION: THE PATIENT HAS BOTH FEVER AND COUGH.

EXPLANATION

THEREFORE: SINCE THE PATIENT HAS BOTH FEVER AND COUGH, THE MOST PROBABLE EXPLANATION IS FLU.

The AI follows the same reasoning process. It starts with the observed symptoms — fever and cough — and evaluates possible reasons using its knowledge base. By ruling out what doesn’t fit, it selects the most probable explanation.

Because different conditions can lead to the same outcome, explanation is not always certain. It is often probabilistic: the best explanation is the one that fits the evidence most strongly, though other explanations may remain possible.

EXAMPLE 3.4 (HI - HUMAN INTELLIGENCE)

Omar sees his front lawn is wet. He considers two possible causes: rain or the sprinkler. Since only the front lawn is wet, he concludes the sprinkler was most likely the cause.

Later, Omar discovers that his children had been playing with water. This new possibility was not part of his original reasoning. With this added information, the sprinkler explanation loses plausibility, and the children's play becomes the best explanation.



EXPLANATIONS CAN CHANGE WHEN NEW POSSIBILITIES ARISE

EXAMPLE 3.4 (AI - ARTIFICIAL INTELLIGENCE)

A medical AI is fed information about a patient who has both fever and cough. From its knowledge base, it considers flu and allergies as possible causes. Since allergies don't usually produce fever, the AI concludes flu is the most probable explanation.

But when the system receives additional information — the patient also has a rash — the plausibility of flu decreases. The new symptom suggests another possible condition, measles, and the explanation shifts to account for this more fitting cause.



CONCLUSION

Inference, decision-making, and explanation all share common ground: they are forms of reasoning that begin with premises — facts, observations, or predictions — and apply a knowledge base to reach conclusions. Each can operate with certainty or with varying degrees of probability.

Where they differ is in direction and purpose. Inference moves directly forward from premises to outcomes, helping us discover new information. Decision-making searches among possible actions to select the one that best meets a goal. Explanation searches backward from an outcome to identify the most plausible cause.

Reasoning takes many forms — whether discovering new information, choosing the best action, or searching for the most plausible cause — it is how we make sense of the world and act within it.

Notes

Common Misconceptions

- *Reasoning outcomes are always certain if the system is well-designed.*
Even with perfect logic, uncertainty arises from incomplete knowledge, noisy data, or ambiguous premises.
- *Decision-making is just picking the highest probability outcome.*
Decision-making is about expected utility, not just probability — the best choice balances both likelihood and value of outcomes.
- *Explanations must be causal.*
In AI, many explanations are associative or model-based rather than strictly causal.
- *One explanation is sufficient.*
Explanation is a search problem: multiple explanations may exist, and plausibility depends on background knowledge and available evidence.

SUMMARY

Reasoning – The structured way we think to discover, decide, or understand.

- Forms of reasoning – inference, decision, and explanation.
- Logic – the rules that guide reasoning.
- Knowledge Base – All forms depend on a knowledge base — rules, facts, or learned patterns that connect conditions to outcomes.

Inference – Arriving at a new piece of information by linking what we already know.

- Starting Point – Premises; can be:
 - Facts – general rules already known.
 - Observations – information detected through our senses.
 - Predictions – expectations based on learned patterns.
- Outcome – New information (a conclusion).
- Strategy – Direct reasoning from premises to outcome (no search).
- Directionality – Forward reasoning.
- Certainty – The certainty of an inference depends on the certainty of its premises.
 - If premises are facts, then inference is definite.
 - If premises are predictions, then inference is probable.

Decision-making – The process of selecting the option with the highest utility (the most useful, valuable, or goal-fitting).

- Starting Point – Premises (facts, rules, observations, predictions, preferences, opinions).
- Outcome – The choice with the highest utility.
- Strategy – Search among possible choices.
- Directionality – Forward reasoning.
- Utility – A measure of how useful, valuable, or goal-fitting an outcome is. It depends on the premises used to judge it.

Explanation – The process of reasoning from a outcome to uncover its causes, or contributing conditions.

- Starting Point – A given (an accepted observation or outcome)..
- Outcome – The reason with the highest plausibility.
- Strategy – Search among possible reasons.
- Directionality – Backward reasoning.
- Utility – A measure of how reasonable or believable an explanation is. It depends on the premises used to support it.

Activity 3.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>T</u>	Reasoning is a structured way of thinking that helps us discover new information, choose actions, or make sense of outcomes.
<u>F</u>	Premises are always absolute facts; observations and predictions cannot be used as premises.
<u>T</u>	The certainty of an inference depends on the certainty of its premises.
<u>F</u>	Inference involves a search among different possibilities before reaching a conclusion.
<u>T</u>	Decision-making is the process of selecting the option with the highest utility.
<u>T</u>	Utility means the usefulness, value, or goal-fit of an outcome.
<u>T</u>	Decision-making and explanation both involve a search strategy, while inference does not.
<u>T</u>	Explanation is reasoning backward from a given outcome to uncover possible conditions or reasons.
<u>T</u>	The plausibility of an explanation depends on how well its premises fit the observed outcome.
<u>F</u>	Explanations are always certain, because once the cause is found it cannot be replaced.

Activity 3.2

Answer the questions below.

1. Reasoning can best be described as:
 - a) Guessing based on experience
 - b) Remembering facts from memory
 - c) **A structured way of thinking that follows rules to reach conclusions**
 - d) Choosing randomly when uncertain

2. In the reasoning process, the starting points we use are called:
 - a) Outcomes
 - b) **Premises**
 - c) Hypotheses
 - d) Causes

3. In reasoning, logic refers to:
 - a) The memory of facts and rules
 - b) The outcome of decision-making
 - c) The search among possible causes
 - d) **The rules that guide reasoning from premises to conclusions**

4. Why is logic important in reasoning?
 - a) It ensures all decisions are fast
 - b) **It provides structure so conclusions follow consistently from premises**
 - c) It allows us to guess without evidence
 - d) It reduces the number of premises needed

5. Which of the following can be premises in inference?
 - a) Facts
 - b) Observations
 - c) Predictions
 - d) **All of the above**

 Activity 3.2 (continued)

Answer the questions below.

6. The certainty of an inference depends on:
 - a) **The certainty of its premises**
 - b) The number of options available
 - c) The directionality of reasoning
 - d) The search process used

7. Decision-making is defined as:
 - a) Selecting the option with the highest plausibility
 - b) Reasoning backward from an outcome to causes
 - c) Directly linking premises to conclusions
 - d) **Selecting the option with the highest utility**

8. In decision-making, utility refers to:
 - a) How probable the outcome is
 - b) **How useful, valuable, or goal-fitting an outcome is**
 - c) How fast the decision can be made
 - d) How certain the premises are

9. Explanation differs from inference mainly because:
 - a) It moves forward from premises to outcomes
 - b) It depends only on facts
 - c) It guarantees certainty
 - d) **It reasons backward from a given outcome to possible conditions**

10. The starting point of explanation is called:
 - a) Premise
 - b) **Given**
 - c) Utility
 - d) Hypothesis

 Activity 3.2 (continued)

Answer the questions below.

11. The plausibility of an explanation depends on:
 - a) The number of possible outcomes
 - b) **The premises that support it and how well they fit the observed outcome**
 - c) The search method used
 - d) Whether the explanation is causal

12. Which of the following requires a search strategy?
 - a) Inference only
 - b) **Decision-making and explanation**
 - c) Explanation only
 - d) Inference, decision-making, and explanation

13. Which of the following best shows inference?
 - a) Deciding which bus to take to school
 - b) Explaining why the lawn is wet
 - c) **Linking a rule and an observation to reach a new conclusion**
 - d) Remembering a fact from memory

14. In decision-making, the best option is chosen by:
 - a) Selecting the one supported by the most facts
 - b) **Selecting the one with the highest utility**
 - c) Selecting the one with the highest probability of occurring
 - d) Selecting the one that requires the fewest premises

15. Why is uncertainty common in explanation?
 - a) Because explanations always require predictions
 - b) **Because multiple possible reasons can fit the same outcome**
 - c) Because causes never align with outcomes
 - d) Because it does not use a knowledge base

Activity 3.3

Read the narrative and answer the questions below.

- a.** In science class, students were learning about photosynthesis. The teacher explained that all plants require sunlight to survive. She then instructed them to start an experiment by placing one plant (A) near the window and another plant (B) inside the closet.

1. Infer the outcome for Plant (A).

Plant (A) will survive.

2. Infer the outcome for Plant (B).

Plant (B) will not survive.

3. What is the general rule you used to make this inference?

All plants require sunlight to survive.

4. State your logic for inferring the outcome for Plant (B).

Since all plants require sunlight to survive, and Plant (B) will not get any sunlight, therefore Plant (B) will not survive.

- b.** An AI model was trained on thousands of cat images to recognize features of cats. From the training data, it learned the general rule that cats usually have pointed ears and whiskers. The model is then shown a new image, which it classifies as a cat.

1. What form of reasoning did the AI model use to make this prediction?

Inference

2. What general rule did it acquire from its training?

If an animal has pointed ears and whiskers, then it is probably a cat.

3. Would you consider this general rule one of certainty?

No, it is not 100% certain. Some breeds of cats do not have pointed ears, so the rule is probabilistic rather than definite.

Activity 3.4

Read the narrative and answer the questions below.

- a. Amir is deciding how to spend his evening. He has two options: study for his math test tomorrow or watch a movie and go to bed early. He knows that studying will help him do better on tomorrow's test, but he also wants to relax after a long day.

1. What form of reasoning is Amir using?

Decision-making

2. What premises is Amir considering to make a choice?

His goal to do well on the test and his preference for relaxation.

3. What outcome does decision-making seek to produce?

The choice with the highest utility.

4. If Amir chooses to study, what does this reveal about his values?

He values doing well on the test more than relaxation.

- b. A delivery drone approaches a tall building on its route. It must decide whether to fly over the building, or around it. Its rules are: (1) minimize delivery time, and (2) conserve battery when possible. The drone maps the routes and calculates that flying over the building will take less time but will require more energy, while flying around it will require more time but less energy.

1. What premises guide the drone's reasoning?

Rules about minimizing delivery time and conserving battery.

2. What other information does the drone need to make its choice?

Which rule is more important in this situation.

3. If the drone chooses to fly around the building, what does this tell you about its programmed priorities?

It assigns higher utility to conserving battery than to minimizing time.

Activity 3.5

Read the narrative and answer the questions below.

- a. Sara walks into the kitchen and sees that the cookie jar is empty. She considers two possibilities: her brother ate the cookies, or her mother gave them to the neighbors. Sara looks around for clues, she notices cookie crumbs on the table.

1. What form of reasoning is Sara using?

Explanation

2. What is the given in this scenario?

The observation that the cookie jar is empty.

3. What possible reasons does Sara consider?

Her brother ate the cookies, or her mother gave them away.

4. Which explanation is most plausible, and why?

The explanation that her brother ate them is most plausible, because the crumbs on the table support this outcome.

- b. A smart home system detects that the living room lights are on in the middle of the night. From past data, it considers two possibilities: someone is awake and moving around, or the timer system malfunctioned. It also detects motion in the living room.

1. What is the given in this scenario?

The observation that the living room lights are on in the middle of the night.

2. What possible reasons does the AI consider?

That a person is awake, or that the timer system malfunctioned.

3. Which explanation is most plausible, and why?

The explanation that a person is awake is most plausible, because the motion sensor confirms activity.

Activity 3.6 - Practicing Reasoning with AI

In this activity, you will interact with an AI which will act as a mentor, providing questions and explanations to deepen your understanding of the topics of the lesson.

In the shared folder you will find five separate text documents, each containing the prompt you need for the activity, as well as the PDF of the lesson summary.

Access the folder from: bit.ly/C4_AI_Activity3_6

a. Reasoning Scenarios

In this part, you will practice three forms of reasoning:

- Inference
- Decision-Making
- Explanation

Each reasoning type has its own text file with a prompt.

Steps to Follow:

1. Download the file from the shared Google folder and choose the prompt for the reasoning type you want to practice.
2. Copy the full prompt and paste it into ChatGPT (or another LLM).
3. The prompt will guide the AI to give you three scenarios. For each scenario, carefully read the setup.
4. Type your answer directly into the chat, always including:
 - Your answer: the outcome, action, or cause.
 - Your reason: why your answer follows from the given rule, priorities, or causes.
5. The AI will confirm if your answer is correct and evaluate whether your reasoning is sound.
6. Continue through the scenarios one at a time until you complete all three.

Tips:

Think about the structure of reasoning:

- Is it predicting? (Inference)
- Is it choosing? (Decision-making)
- Is it explaining? (Explanation)

Always explain your reasoning, even if unsure — that is what the AI will evaluate.

Don't rush; reflect on why a certain answer follows from the premises.

b. Questions & Answers

In this part, you will test your understanding of the lesson using the PDF provided in the folder. You can choose between:

- True/False Questions
- Multiple-Choice Questions

Each activity has its own text file with a prompt.

Steps to Follow:

1. Open the shared folder and download the PDF of the lesson summary.
2. Attach the PDF in ChatGPT.
3. Copy the prompt for the activity you want to practice (True/False or Multiple Choice) and paste it into ChatGPT.
4. The AI will create 10 questions based on the summary:
 - If you chose True/False, each question will be answered with “True” or “False.”.
 - If you chose Multiple Choice, each question will have 4 options (A, B, C, D).
5. For each question:
 - Read carefully and submit your answer.
 - The AI will tell you if you are correct, provide the correct answer if needed, and give a short explanation.
6. After all questions:
 - The AI will list the questions you got wrong.
 - It will give you a short review of the key topics you struggled with.
 - It will suggest a follow-up exercise to strengthen your understanding.

Tips:

- For True/False: think about why the statement is true or false before answering.
- For Multiple Choice: read all four options carefully before choosing.
- Use the AI’s feedback to focus on the concepts you find most difficult.
- Repeat the activity for more practice — the AI can generate new sets of questions each time.